

AMAN KUMAR PRAJAPATI

+91 8306711315 [LinkedIn](#) [GitHub](#) 22bec006@iiitdwd.ac.in

EDUCATION

Indian Institute of Information Technology (IIIT) Dharwad

B.Tech in Electronics and Communication Engineering, CGPA: 9.42/10

Dharwad, Karnataka, India

2022 - 2026

Indian Institute of Technology (IIT) Madras

Diploma in Data Science and Applications

Mode - Online

2024 - 2025

RESEARCH AND WORK EXPERIENCE

U R Rao Satellite Centre, ISRO

Research Intern

May 2025 – Jun 2025

Bangalore, India

- Built a *physics-informed autoencoder* (PIAE) that embeds rigid-body dynamics in the loss, denoises DTG angular-rate telemetry, and flags non-physical spikes.
- Integrated the PIAE with cyclic-redundancy checks (CRC) to form a hybrid anomaly-detection pipeline that slashes false alarms while preserving time-to-detection.
- Benchmarked LSTM, Transformer, Isolation Forest, and one-class SVM baselines on DTG flight + synthetic data, standardising evaluation with Precision, Recall, and F1.

Strative.ai

ML Engineer Intern

Aug 2024 – Oct 2024

Remote

- Developed RAG pipeline using python, LangChain, Llama 3, Elasticsearch, Milvus database; performed evaluation using RAGAS.
- Implemented a blended RAG with dense & sparse embeddings with hybrid query strategies; achieved 89% top-10 retrieval accuracy on NQ dataset, beating current benchmarks; extended experiments to Trec-covid & SQuAD datasets, used metrics Exact Match & nDCG@k.
- Used Ray.io in GKE for auto scaling, bulk-ingestion & multi-search, reducing data ingestion, retrieval & evaluation time by 80%.

L&T Technological Services

ML Engineer Intern

May 2024 – July 2024

Bangalore, India

- Fine-tuned YOLOv5 and RT-DETR on **US railway tracks dataset** for real-time **crack detection & classification**.
- Achieved **86% precision**; improved compute time by **68%** through **OpenMPI parallel processing**.
- Implemented secure **WireGuard server** for stable communication and file exchange with **40+ remote devices** internationally.
- **Tech:** Python, OpenCV, PyTorch, CUDA, OpenVINO

Humors Tech (Health-tech Start-up)

Research Intern

Apr 2024 – Aug 2024

Remote

- Fetched raw breath data via the company API and converted the samples to standard units for the *Respyr* product.
- Built and trained RNN/LSTM models in PyTorch to classify lung health as *Poor*, *Normal*, or *Good*, achieving a 92% F1 score.
- Developed a Flask dashboard that shows users their real-time lung-health status and model insights.
- **Tools:** Python, Flask, Matplotlib, Seaborn, Pandas, scikitlearn, Keras

Speech Processing Lab, Indian Institute of Information Technology (IIIT)

NLP Research Intern

Aug 2023 – Nov 2023

Dharwad, India

- Worked with a small team to build a real-time English/Hindi ↔ Kannada translation model and deploy (“burn”) it onto custom embedded hardware for on-device inference.
- Trained, fine-tuned, and quantized multiple models (Transformer and RNN variants), then evaluated them on an English to Kannada and Hindi to Kannada parallel corpus—best model reached 87% BLEU.
- Owned the data pipeline: collected raw text, removed noise, aligned sentence pairs, tokenized, and generated train/validation /test splits prior to model training.

TEACHING EXPERIENCE

Atal Innovation Lab, Mallasajjan English Medium School Dharwad

Undergraduate Teaching Assistant

Oct 2023 – Nov 2024

Dharwad, Karnataka, India

- Instructed high-school students in the fundamentals of embedded electronics, covering microcontrollers (Arduino, ESP32), sensors, basic C/C++, Python programming, and circuit-building best practices.
- Designed and supervised hands-on projects such as smart-home automation modules, line-following robots, and IoT weather stations, guiding students from schematic to prototype and final demo.
- Taught 3D-model creation in Fusion 360, including parametric design, slicing, and entry-level 3D-printing workflows; students produced custom enclosures for their electronics projects.
- Served as hackathon mentor for the school teams for nationwide competitions on ideation, rapid prototyping, and presentation skills; several teams finished in the top 10% nationally.

PUBLICATION

Compress, Encode, Diagnose: PCA-Enhanced Quantum Neural Networks for Breast Cancer Classification <i>IEEE RASSE 2025</i>	<i>In review</i>
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PROJECTS

Autonomous Navigation on Indian Roads (Ongoing) <i>Research</i> ■ Building a multimodal dataset of 120 km driving on Indian highways and city streets (30 FPS front-facing video + sensor streams). ■ Designing a 3D-CNN + sensor-transformer model that forecasts vehicle speed and lateral path.
Multimodal LLM-based Recommendation System ■ Fine-tuned LLaVA on a custom leaf-disease image/text set, reaching 90% accuracy in multi-class detection. ■ Benchmarked LLM dependability via metric-driven experiments, ensuring reproducible results across prompt variants. ■ Launched an AI chatbot that now resolves 75% of routine farmer inquiries autonomously, cutting response time from hours to minutes. ■ Tech: Python, Langflow, LLaVA
AI-Powered Smart Agriculture Platform ■ Designed and built an automated smart agriculture drone integrated with ESP32 for flight control and Raspberry Pi 4 for real-time NDVI-based crop health monitoring, achieving 24% improved yield prediction accuracy and 50% reduction in manual inspection time. ■ Deployed edge computing on Raspberry Pi 4, running locally hosted LLM-guided systems for irrigation scheduling and disease detection, resulting in 30% less water usage and 40% lower latency. ■ Developed a live web dashboard (HTML/Tailwind/JS) for real-time visualization of drone-captured sensor data, enhancing decision-making speed by 35%. ■ Tech Stack: Python (TensorFlow, OpenCV), Raspberry Pi 4, ESP32 (C/C++ , SPI), SSH, Pi NoIR Camera, DHT11, Soil-Moisture Sensors.

PROFESSIONAL & EXTRA-CURRICULAR ACTIVITIES

Vice President	EPOCH – Data Science and Artificial Intelligence Society, IIIT Dharwad	Jan 2024 – Present
Head, IoT Group	IRIS Innovation & Robotics Club, IIIT Dharwad	Jan 2024 – Present
Vice President	AccelAirate Club, IIIT Dharwad	Jan 2024 – Present
Core Member	Google Developer Student Club, IIIT Dharwad	Aug 2023 – Present
Guitarist	440hz Music Band, IIIT Dharwad	Mar 2023 – Present
Cinematographer	Iridescence Film & Photography Club, IIIT Dharwad	Mar 2023 – Present

AWARDS & FELLOWSHIPS

IASc–INSA–NASI Summer Research Fellow <i>Indian Science Academies (IASc, INSA, NASI)</i>	2025
CHANAKYA Fellowship <i>Technology Innovation Hub on IoT, IIT Bombay</i>	2024
Winner (1st Place) — Clash of Data Analysts (CODA), 48-hour Hackathon <i>IIIT Dharwad</i>	2023
Gold Medal — National Science Olympiad (NSO) <i>Science Olympiad Foundation (SOF)</i>	2015

TECHNICAL SKILLS

Languages	Python, C++ , MATLAB, R, SQL, HTML, CSS
ML frameworks	PyTorch, TensorFlow
Technologies	Linux, Shell Scripting, Version control (Git)
Software	Microsoft Office, PyCharm, Excel , Autodesk Fusion, Docker
Packages	NumPy, Pandas, SciPy, TensorFlow, PyTorch, OpenCV, scikit-learn, Statsmodels, Matplotlib
Analytical Skills	Regression, Classification, Clustering, Neural Networks, Data Analysis

CERTIFICATIONS

2024	Generative AI with Diffusion Models (Nvidia)	Certificate
2024	Deep Learning Foundation (Kaggle)	Certificate
2024	Machine Learning Foundation (Kaggle)	Certificate
2023	Google Cloud Certified - Cloud Computing Foundations and Generative AI	Certificate
2023	Foundation level in Programming and Data Science (IIT MADRAS)	Certificate